

PREPROCESSING METHOD FOR DCT-BASED IMAGE COMPRESSION WITH ADAPTIVE THRESHOLDING

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ABSTRACT

In DCT-based image compression standards such as JPEG, visible ringing artifacts are a common issue, particularly at low bit rates. While applying a global low-pass filter during preprocessing can reduce these artifacts, it inevitably blurs the entire image and degrades visual quality. Existing selective preprocessing methods attempt to mitigate this by filtering only regions with strong edges based on a fixed auto-correlation threshold. In this paper, we propose an optimized preprocessing method that introduces an Adaptive Thresholding mechanism. By analyzing the global variance of the input image, our method dynamically adjusts the filtering threshold to suit the image's complexity. Experimental results demonstrate that our adaptive approach outperforms both global filtering and fixed-threshold methods in terms of SSIM, FSIM, and VIF metrics, effectively suppressing ringing artifacts while preserving texture details.

Index Terms— DCT, Image Compression, Ringing Artifacts, Adaptive Threshold, Preprocessing

1. INTRODUCTION

DCT-based image compression, such as JPEG, often suffers from visible ringing artifacts around sharp edges at low bit rates due to the Gibbs phenomenon. While global low-pass filtering can mitigate these artifacts, it inevitably causes excessive blurring. Oizumi et al. [1] addressed this with a selective preprocessing method using a fixed auto-correlation threshold; however, a fixed value lacks the flexibility to adapt to images with varying complexity. In this paper, we propose an **Adaptive Threshold** mechanism that dynamically adjusts the filtering threshold based on the global variance of the input image. This approach ensures that complex textures are preserved while smooth regions are effectively filtered, achieving a superior balance between artifact suppression and detail preservation.

2. RELATED WORK

2.1. Pre-Processing Methods

Some methods attempt to filter the input image before compression to avoid producing severe artifacts. However, simple linear filtering often causes loss of fine detail and over-smoothing. Oizumi's method [1] improves upon this by using local statistics (auto-correlation) to selectively filter edges.

2.2. Post-Processing Methods

Many researchers have proposed artifact removal filters applied after decompression. However, these methods require modifications on the decoder side, which is often not compatible with existing hardware or standard JPEG/MPEG decoders. Our approach focuses on the encoder side (preprocessing), ensuring full compatibility with standard decoders.

3. METHODOLOGY

3.1. Selective Preprocessing Algorithm

The core algorithm is based on the observation that ringing artifacts are most visible around strong edges. The process involves:

1. **Auto-correlation Calculation:** For each pixel, a modified auto-correlation coefficient (ρ_{mod}) is calculated using its local neighborhood. A high positive value indicates a strong edge.
2. **Thresholding:** The coefficient is compared against a threshold (ρ_{th}). If $\rho_{mod} > \rho_{th}$, a low-pass filter is applied.

3.2. Adaptive Threshold Optimization

In the original method [1], ρ_{th} is a fixed value (typically 0.3). We propose that complex images (high variance) require a higher threshold to avoid blurring real textures, while simple images (low variance) can tolerate a lower threshold.

We define the adaptive threshold $\rho_{adaptive}$ as a function of the image's global variance (Var_{global}). In our implementation, we calculate the variance of the Y channel. We use linear interpolation to map the variance range [500, 5000] to a threshold range [0.2, 0.45].

$$\rho_{adaptive} = \text{interp}(Var_{global}, [500, 5000], [0.2, 0.45]) \quad (1)$$

For high-variance images, the threshold increases to protect textures. For low-variance images, the threshold decreases to aggressively suppress ringing in flat areas.

4. EXPERIMENTAL RESULTS

We implemented the proposed method in Python and compared it against three other schemes: Direct Compression (no preprocessing), Global Filtering, and the Fixed Threshold method [1]. The test images were compressed using JPEG with a quality factor of 10.

4.1. Quantitative Comparison

We evaluated the performance using Peak Signal-to-Noise Ratio (PSNR), Structural Similarity (SSIM), Feature Similarity (FSIM) [2], and Visual Information Fidelity (VIF) [3].

Table 1 shows the results for the test image. As shown, our **Adaptive Threshold** method outperforms both the Global Filtering and Fixed Threshold methods across SSIM, FSIM, and VIF metrics. Specifically, the SSIM improved from 0.7868 (Fixed) to 0.8045 (Adaptive).

Table 1. Comparison of Preprocessing Methods (JPEG Q=10)

Method	PSNR	SSIM	FSIM	VIF
Direct Comp.	26.01	0.8369	0.8844	0.3624
Global Filter	23.81	0.7303	0.6980	0.2733
Fixed Thres. [1]	24.85	0.7868	0.8043	0.3106
Adaptive (Ours)	25.23	0.8045	0.8334	0.3258

4.2. Visual Quality

Visual inspection confirms that Global Filtering results in excessive blurring. Fig. 1 demonstrates the ringing artifact suppression. The left side shows the ringing artifacts present in standard compression, while the right side shows our result with reduced artifacts.

Additionally, Fig. 2 compares the texture preservation capabilities. The Global Filtering method (Right) causes significant blurring in the detailed regions (highlighted in red

circles). In contrast, our Adaptive Threshold method (Middle) successfully preserves these textures, resulting in a visual quality that is much closer to the Original Image (Left).



Fig. 1. Visual comparison of ringing artifact suppression. Left: Only Compression (showing ringing). Right: Compression with preprocessing Adaptive Threshold (reduced ringing).

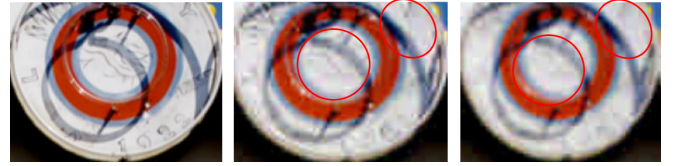


Fig. 2. Comparison of detail preservation. Left: Original image. Middle: Compression with preprocessing Adaptive Threshold. Right: Global Filtering. Note how Global Filtering blurs the texture, while our Adaptive method maintains sharpness.

5. CONCLUSION

This paper presented a preprocessing method for DCT-based image compression utilizing an adaptive thresholding strategy. By dynamically adjusting the filtering threshold based on the global variance of the image, we improved the trade-off between ringing artifact suppression and texture preservation. Experimental results confirm that the proposed adaptive method yields higher objective quality metrics compared to the conventional fixed-threshold approach.

6. REFERENCES

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